



Crop pest image recognition based on the improved ViT method

Xueqian Fu ^{a,b,c,d,*}, Qiaoyu Ma ^{a,b}, Feifei Yang ^a, Chunyu Zhang ^{a,b}, Xiaolong Zhao ^a, Fuhao Chang ^a, Lingling Han ^a

^a College of Information and Electrical Engineering, China Agricultural University, Beijing, China

^b National Innovation Center for Digital Fishery, China Agricultural University, Beijing, China

^c Key Laboratory of Smart Farming Technologies for Aquatic Animal and Livestock, Ministry of Agriculture and Rural Affairs, Beijing, China

^d Beijing Engineering and Technology Research Center for Internet of Things in Agriculture, Beijing, China

ARTICLE INFO

Article history:

Received 6 January 2023

Received in revised form

2 February 2023

Accepted 13 February 2023

Available online 18 February 2023

Keywords:

Crop diseases recognition

Fruit tree leaf disease

Vision Transformer

Deep learning

Computer Vision

ABSTRACT

The crop pests and diseases in agriculture is one of the most important reason for the reduction of bulk grain and oil crops and the decline of fruit and vegetable crop quality, which threaten macroeconomic stability and sustainable development. However, the recognition method based on manual and instruments has been unable to meet the needs of scientific research and production due to its strong subjectivity and low efficiency. The recognition method based on pattern recognition and deep learning can automatically fit image features, and use features to classify and predict images. This study introduced the improved Vision Transformer (ViT) method for crop pest image recognition. Among them, the region with the most obvious features can be effectively selected by block partition. The self-attention mechanism of the transformer can better excavate the special solution that is not an obvious lesion area. In the experiment, data with 7 classes of examples are used for verification. It can be illustrated from results that this method has high accuracy and can give full play to the advantages of image processing and recognition technology, accurately judge the crop diseases and pests category, provide method reference for agricultural diseases and pests identification research, and further optimize the crop diseases and pests control work for agricultural workers in need.

© 2023 China Agricultural University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Crop diseases and insect pests have a great impact on the yield reduction of staple grain and oil crops, and the quality

decline of fruit and vegetable crops. Crop pests and diseases seriously affect regional food production and ecological security, and then threaten macroeconomic stability and sustainable development. At present, pests and diseases have reduced the potential crop yield by an average of 40 % all over the world, causing losses of about one-tenth of the world's total grain production. According to statistics, in 2020, China's cumulative area of major crop diseases and insect pests reached 300 million hm². In recent years, with the significant improvement of China's plant protection capacity, through

* Corresponding author. Address: CIEE 633, 17 Qinghua Donglu, Beijing 100083, People's Republic of China.

E-mail address: fuxueqian@cau.edu.cn (X. Fu).

Peer review under responsibility of Peer review under the responsibility of China Agricultural University.

<https://doi.org/10.1016/j.inpa.2023.02.007>

2214-3173 © 2023 China Agricultural University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

the adoption of various pest prevention and control measures, China can reduce grain loss by 870–110 million tons annually, accounting for 16 % – 19.55 % of the total national grain output [1]. Therefore, the key link to improving crop yield and ensuring quality and safety is how quickly and accurately identify crop diseases and pests to prevent and control them. However, traditional manual recognition and expert systems do not meet the requirements of agricultural modernization because of factors such as accuracy and response speed. Image recognition combined with artificial intelligence technologies such as statistical machine learning [2,3] and deep learning [4] for monitoring and classifying crop pests has the advantages of high accuracy, good real-time, and non-destructive, which can provide more effective technical provisions for the prevention and control of agricultural pests [5]. It has become a key technology to assist agricultural security early warning and a research hotspot in related fields, which is significant for the development of smart agriculture and agricultural energy internet [6,7].

The manifestations of diseases and pests is regional in space, and random, sudden and uncertain in disease form, which makes it difficult to accurately predict, identify and control [8]. The image recognition methods of crop diseases and pests mainly include the following categories: (1) Based on manual recognition methods, early crop diseases and pests are recognized mainly through manual on-site inspection, measurement, statistics and identification. Traditional manual identification methods require agricultural producers to master a large amount of accurate crop growth information, analyze the damage degree of crop diseases and insect pests, and can no longer satisfy the need of scientific agricultural production and disease prevention in terms of identification efficiency, identification accuracy, application scenarios, etc. The main limitation of manual recognition methods is strong subjectivity and low efficiency [9]. (2) The second is an instrument-based identification method, which uses various portable crop pest detectors. It was proved that conventional gel electrophoresis and TaqMan probe can quickly confirm thousand cankers disease from lesion samples [10]. The limitation of the instrument-based identification method is that it still needs a manual operation, and the degree of automation and intelligence is low. (3) The third category is recognition methods based on pattern recognition and machine learning [11]. Pattern recognition is to learn the sample characteristics on the basis of a certain mathematics and calculation, and divide the sample to be recognized into the category. The unknown data can be predicted by automatically acquiring rules from the known data using machine learning.

With the development of artificial intelligence identification technology in agricultural pest identification systems, artificial intelligence identification has become the most promising pest identification method due to its fast speed, good stability and high accuracy [12]. Kale et al. detected diseases using random forest (RF), support vector machine (SVM), and K-nearest neighbor (KNN) [13]. It was found that a multivariate support vector machine was accurate for classification to detect crop diseases. The main problem in its application was that it was difficult to extract features. Gowrishankar et al. detected peanut leaf diseases and pests using

the binomial logistic regression and support vector machine (BLR-SVM) model and evaluated its efficiency [14]. Raza et al. used SVM to monitor tomato powdery mildew in combination with infrared image information and depth information [15]. The method of using machine learning to identify crop pests has poor robustness and low recognition accuracy. Deep learning has the ability to automatically extract image features with strong generalization ability, showing excellent performance in the identification of crop pests [16]. Wang et al. presented the sampling balanced region proposal network (S-RPN) to monitor crop pests combining the attention mechanism and the residual network. The average recall rate was superior to the multi-box detector, RetinaNet and other methods [17]. Tassis et al. integrated multiple convolutional neural networks to automatically detect coffee crop pests in combination with instance and semantic segmentation [18]. The research results showed that it was suitable to be implemented in the embedded platform for application. With the deep development and integration of modern information technology and agriculture, “smart agriculture” relies on image acquisition equipment, sensors and other terminals to build an Internet of Things system [19]. On the basis of a large amount of normal and pest image data in real-time, smart agriculture adopts deep learning to identify crop pests and diseases, which is characterized by rapidity, accuracy, real-time, etc. It is an inevitable trend for future development of smart agriculture to combine deep learning with image processing technology to identify crop pests.

The accurate recognition of crop pest images is still a challenging problem. On the one hand, the complex background, changes in lighting conditions, overlap and occlusion, especially the close colour of foreground objects and background leaves, bring great difficulties to target recognition. The traditional machine learning algorithm cannot extract and detect the image recognition features related to diseases and pests from the complex and changeable background environment, and the established crop diseases and pest recognition model is not stable [20]. On the other hand, the existing pest image samples are relatively small, the collection of pest images in the field is difficult and the workload is also large. At the same time, the collected pest images may also have the problem of missing important features. How to use a small number of existing pest image depth mining image features to improve the accuracy of pest image recognition has become an urgent technical problem.

In this study, apple and grape leaves are used as research objects. Based on the improved ViT method, combined with block division, transformer and attention mechanism, a crop pest image recognition algorithm is proposed. The apple (scab), apple (black rot), apple (cesar apple rust), apple (health), grade (black rot), grade (leaf light), and grade (health) were classified and identified. The proposed model can give full play to the advantages of image processing and recognition technology, and quickly judge the damage degree of crop diseases and pests. The specific innovation work is as follows: (1) We use the block coding method of ViT to extract spatial features from the data. This strategy can effectively select the region with the most obvious features. (2) We propose a method based on the improved ViT model, which uses a transformer as the attention mechanism to deeply mine the

relationship between image features in each patch, and can better mine the accuracy of pest features.

2. Materials and methods

2.1. Technical route

The software consists of two parts: the classifier training module and the classifier test module. It uses the Partition algorithm, Transformer, and Attention Mechanism. The idea is as follows: In the classifier training module, first, convert the RGB image in the training image library to a uniform size. Second, the original images were divided into N non-overlapping patches, using the block encoding idea of ViT, to segment the images and extract the spatial features of the data. Thirdly, the relationship between image features in each patch is deeply mined by Transformer. Finally, the output layer of the model uses the softmax classifier and finally calculates the category probability to determine the corresponding pest level of the leaves. In the classifier test module, test images are used to test the classifier training results, and the test indicators use recall and accuracy. The technical route of the classifier training module is shown in Fig. 1.

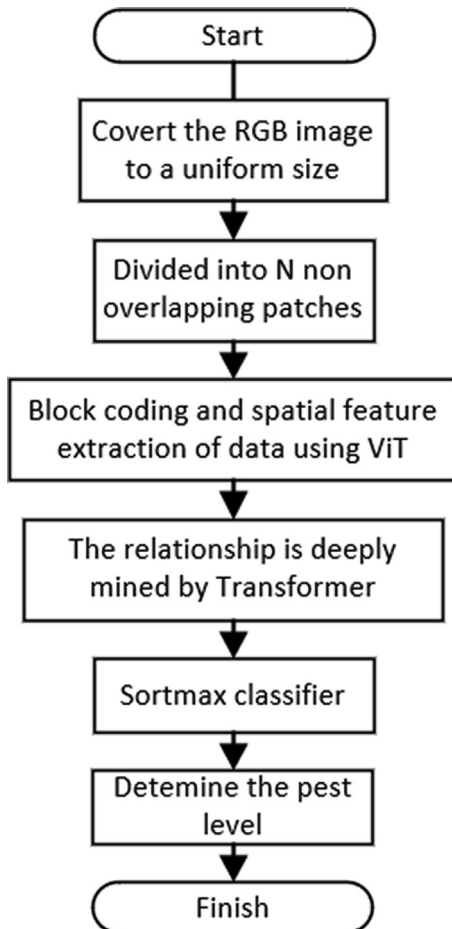


Fig. 1 – Technical route of classifier.

2.2. ViT

Compared to convolutional neural networks (CNNs), the Vision Transformer (ViT) relies on excellent modeling capabilities to achieve very good performance on several benchmarks such as ImageNet, COCO, and ADE20k [21]. ViT is inspired by the self-attention mechanism in natural language processing, where word embeddings are replaced with patch embeddings. ViT divides the input image into 16×16 , and applies a standard Transformer to extract image features. Fig. 2 illustrates the model structure of the ViT.

In the Patch Embedding phase, images are divided into fixed-size patches. Each image will generate n patches, that is, the input sequence length is n . Then it is converted into n tokens through linear mapping. A special character cls needs to be added here, so the final dimension is $n + 1$. In order to preserve the position information of the sequence, Positional Encoding needs to be added. The location code can be understood as a table with n rows in total. The size of n is the same as the length of the input sequence. Each row represents a vector, and the dimension of the vector is the same as that of the input sequence code.

After the input sequence processed above is input into Transformer Encoder, the dimension of the output sequence is adjusted by MLP Head to make the output dimension consistent with that before input into Transformer. Finally, we get the classification results through Classifier.

2.3. Transformer

Transformer is the core part of ViT. Unlike RNN, Transformer can be trained in parallel. Transformer itself cannot use the sequence information of the sequence, so you need to add the position Embedding in the input, otherwise, Transformer is a bag of words model. The main part of Transformer is the Self-Attention structure, where the used Q , K , and V matrices are obtained by the linear transformation of the output. The main part of Transformer is the Self-Attention structure, where the used Q , K and V matrices used are obtained by linear transformation through the output. Multiple Self-Attention in Multi-Head Attention in Transformer can capture the correlation coefficients in multiple dimensions between words.

The transformer contains an encoder and a decoder. The encoder part maps the input sequence $x=(X_1, \dots, X_n)$ represented by symbols to the sequence represented continuously $z=(z_1, \dots, z_n)$. For a given sequence z , the decoder generates a set of output sequence $y=(y_1, \dots, y_m)$ at a time. In each step, the model is automatically regressed. When generating the next set of output sequences, the previously generated output sequences are used as additional information inputs. The overall structure of the Transformer is composed of L same layers. Each layer consists of Multi-Head Self-Attention (Eq. (2)) and a simple Full Connection Feedforward Neural Network (Eq. (3)). Fig. 3 illustrates the model structure of the Transformer.

$$z_0 = [x_{\text{class}}; x_p^1 E; x_p^2 E; \dots; x_p^n E] + E_{\text{pos}} \quad (1)$$

$$z'_l = \text{MSA}(\text{LN}(z_{l-1})) + z_{l-1}, l \in [1, L] \quad (2)$$

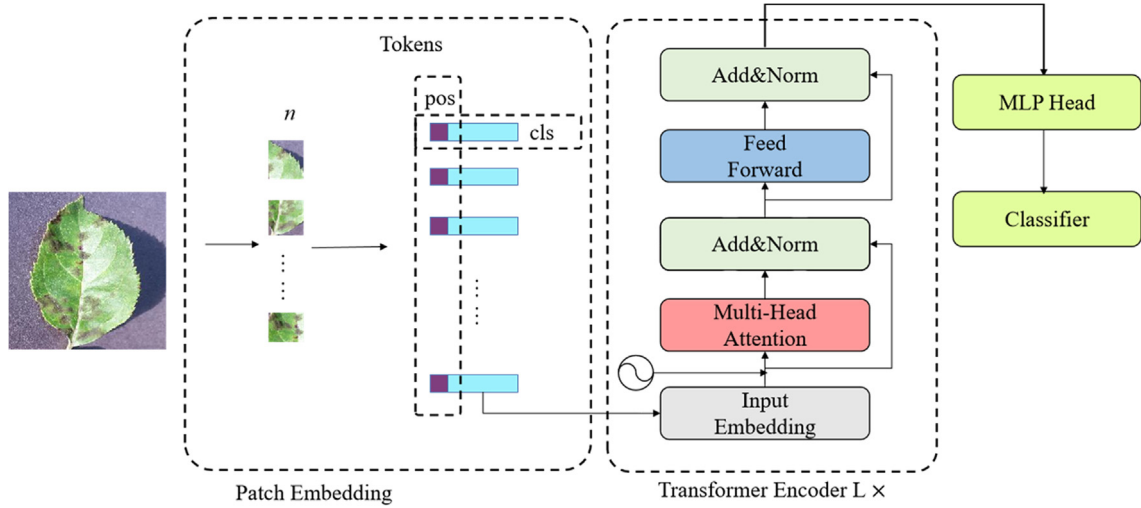


Fig. 2 – Structure of model.

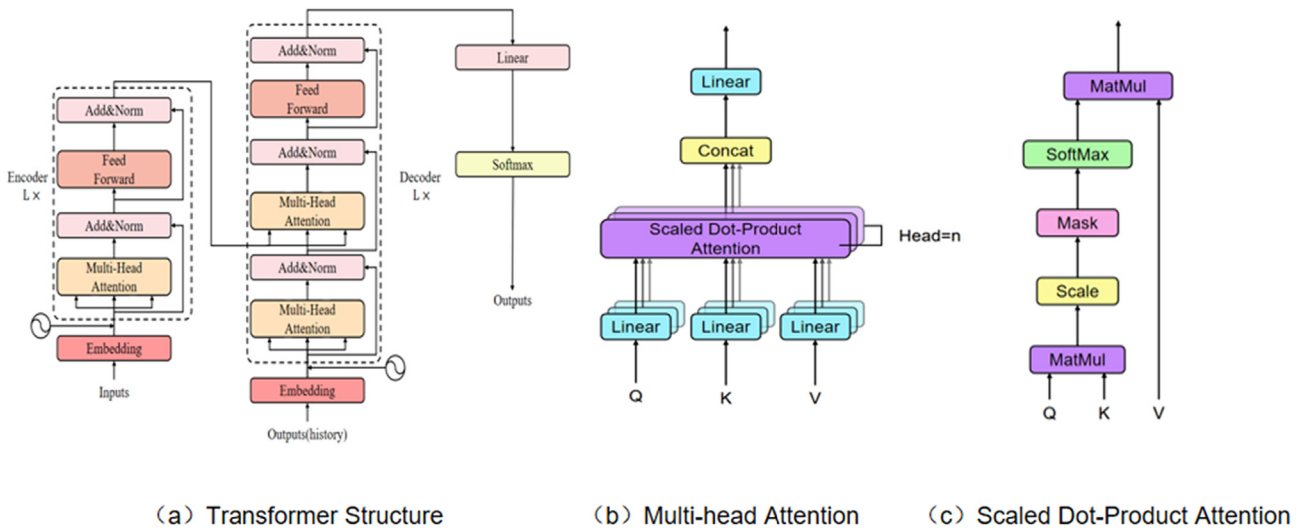


Fig. 3 – Structure of the Transformer. (a) Multi-head attention and MLP stacking are implemented in Transformer. **(b)** Multi-headed attention splits the input data into multiple parts, which are calculated separately using self-focused attention. **(c)** Self-attention is realized by matrix multiplication.

$$z_l = \text{MLP}(\text{LN}(z_l^0)) + z_l^0, l \in [1, L] \quad (3)$$

$$y = \text{LN}(z_L^0) \quad (4)$$

where, x_{class} is Special character vector, $x_p^1 \dots x_p^n$ is input sequence obtained from divided images, E is a matrix for linear mapping, E_{pos} is a matrix for positional encoding, z_0 a continuous sequence to be input into encoder, z_l^0 and z_l are the outputs of each layer of their respective corresponding networks, y represents the layer, z_L^0 is the output of last layer, y is a set of output sequences generated by the decoder, L is the number of layers, LN is normalization operation.

Sometimes we expect the network to see the overall situation, but also focus on key information. For example, in nat-

ural language processing, a word in a sentence is often not independent of its context, but its relevance is different from that of a word in the context. Therefore, when we process this word, we should focus more on the words with higher relevance while seeing its context, which requires the use of the often-said self-attention mechanism. For the sequence of pictures, the position of pictures before the division is also important. However, it has a disadvantage: when the model encodes the current position information, it will excessively focus on its own position. Therefore, it is proposed to solve this problem through multiple attention mechanisms.

Multi-Head attention in the transformer needs to splice n times of Scaled Dot Product Attention (SDPA) (Eqs. (7)–(8)) results. The final calculation result is obtained by a linear

transformation of the spliced results. Fig. 2 (c) is the structure diagram of SDPA. Q , K , V in the figure represent the projections of sequences in different spaces. The point product is used for similarity calculation to increase the expression ability of attention weight on the original sequence. Scaled Dot Product Attention outputs the weighted value of V . The weight assigned to each V is calculated by Q and K .

$$X = \text{LayerNorm}(X + \text{MultiHead}(X)) \quad (5)$$

$$X = \text{LayerNorm}(X + \text{FeedForward}(X)) \quad (6)$$

$$\text{where } \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (7)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

where, X is the feature to be processed, LayerNorm is a normalized layer, Q is Query, K is Key, V is Value, and d_k is the corresponding parameter.

The Multi-Head Attention mechanism divides sequence data into multiple sequences, extracts sequence information of different parts through MLP structure, and represents features from different subspaces (Eqs. (4) and (5)). In this way, each head can cut down the number of features that the model needs to consider when calculating, and the total calculation cost is significantly reduced compared with the Single-head Attention to all dimensions. The MLP structure in the Transformer consists of two projections of the GELU nonlinear separation activation function. The marking dimension d of the input data is fixed in all layers. Finally, a linear projection is used to classify the coded inputs.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^o \quad (9)$$

$$\text{where } \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (10)$$

where, the parameter matrices used for projection are $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, $W^o \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

2.4. Dataset

The dataset is from the crop leaf image data provided under the topic of Smart Agriculture Pest and Disease Precise Identification in the Beijing Big Data Skills Competition. The com-

petition aims to build a pest identification model, accurately identify crops and pest types, provide reasonable suggestions for farmers to actively prevent pests, and solve the problem of pest identification in smart orchards.

It has a total sample size of 4562. In the experiment, it is divided into a training set and a test set. The number of training set samples is 3581, and the number of test set samples is 981. Tags of the apple sample include: scab, black rot, cedar apple rust, healthy (see Fig. 4). Tags of the grape sample include: black rot, leaf light and healthy (see Fig. 5). In the training set, the number of samples corresponding to the above tags is 375, 371, 165, 987, 705, 628, 250 respectively. In the test set, the number of samples corresponding to the above tags is 125, 123, 55, 328, 58, 209 and 83 respectively.

Among them, apple scab is an important disease in all apple producing areas in the world. It will cause apple quality, yield and storage, and reduce the proportion of commercial fruits. Apple scab control should adhere to the principle of prevention first and treatment second, so as to effectively control the occurrence of the disease and reduce the harm. Rust fungi can cause rust in crops. Among the diseases of apples, the frequency of cedar apple rust is relatively high. This rust causes parts of the crop to turn yellow. It is named because the shape of this lesion location resembles rust. Leaf blight in grapes causes disease spots on the leaves of grapes, which usually vary in size and brown color. As the infection increases, the spots gradually become larger and connected, and the color gradually darkens. The dry area reaches 1/3–1/2 of the leaves. There is a zone deeper than the disease spot at the edge of the disease spot; The boundary between disease and health is obvious. Black rot is a disease that can occur during the growth of both apples and grapes. In the early stages, we will see that their metamorphosis lesions are brown. As the degree of lesion deepens, it spreads to various locations of the fruit. When the fruit changes color in various places, the fruit will lesion black, and at the same time, the fruit will not soften.

The dataset that will play a role in this paper includes several common pests and diseases mentioned above. These diseases can be observed by human eyes, which is beneficial to experimenters intuitively feeling the training effect of the current model through observation, and facilitate further improvement and adjustment.

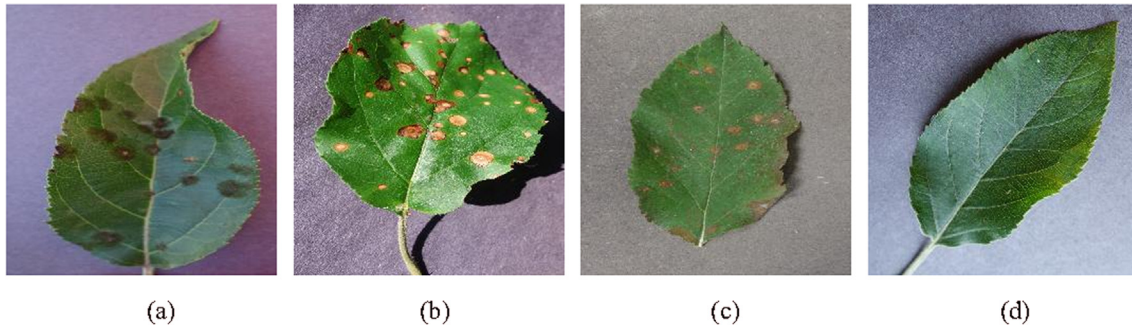


Fig. 4 – Example of apple leaf images: (a) scab; (b) black rot; (c) cedar apple rust; (d) healthy.

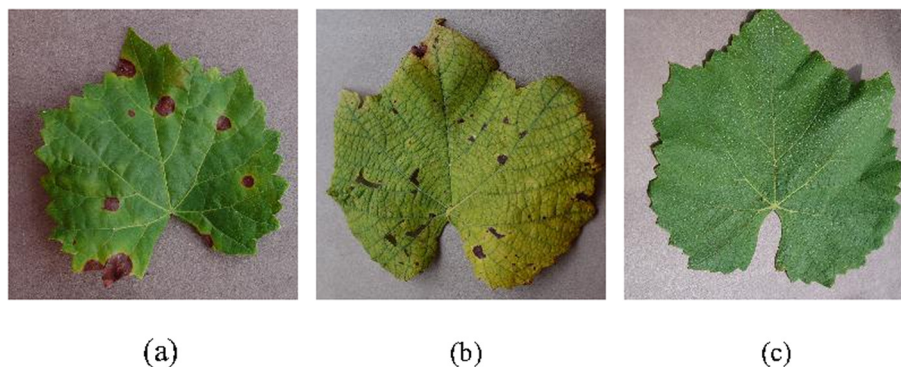


Fig. 5 – Example of grape leaf images: (a) black rot; (b) leaf blight; (c) healthy.

2.5. Metric

In AI, especially in classification tasks, confusion matrix can directly reflect the classification results of the model. Its presentation form is a table, which can know the prediction results of each category item from the vertical view and the actual category of each category item from the horizontal view. At the same time, we can use this tool to easily get the values of some indicators, such as *precision* (Eq. (11)) and *recall* (Eq. (12)).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

where, *TP* stands for positive sample that is judged to be correct; *FP* stands for positive sample of misjudgment; *FN* represents positive samples that are incorrectly predicted to be negative.

We need to further understand precision and recall. The former focuses on predicting the correct results, considering how many of the correct samples are judged to be positive. The latter focuses on the whole sample itself, considering how likely it is to be predicted if a sample is positive.

The operating environment of the system used in this experiment needs to meet the following requirements: Intel or AMD supports 64-bit processors, no less than 16 GB memory and 100 GB disks, no less than 3 GB video memory, Windows operating system, and Python with version 3.8 or higher.

3. Results and discussion

3.1. Result

The confusion matrix is one of the indicators for model classification and detection evaluation. During the model training process, the confusion matrix is obtained as shown in Fig. 6. The row label of the confusion matrix represents the real cat-

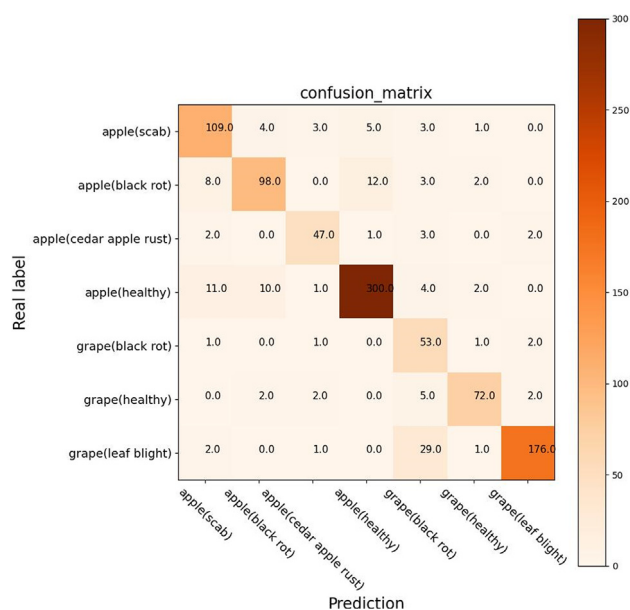


Fig. 6 – The confusion matrix of the adopted model on the test set.

egory, and the column label indicates the predicted category. Only when the row and column labels are the same, it means that the model prediction is correct, otherwise, it means that the model prediction is wrong, and the sum of the corresponding row values is used as the total number of samples of this category. The darker the corner line, the better the effect of the model. In this experiment, apple (scab) selects 125 pictures, apple (black rot) selects 123 pictures, apple (cedar apple rust) selects 55 pictures, apple (healthy) selects 328 pictures, grape (black rot) selects 58 pictures, grape (healthy) selects 83 pictures, and grape(leaf blight) selects 209 pictures to test its accuracy.

Take 109 in row 1 and column 1 of the data as an example. Looking vertically, it means that among the 133 samples classified as apple(scab) by the software, the number of real labels as apple(scab) is 109. Through these two data, we can get the recognition accuracy rate of the software for the category label of apple (scab). The result is a ratio of 109 to 133, which is 0.82; Horizontally, it shows that among the 125 samples whose true labels are apple(scab), the number of samples recognized by the software as apple(scab) is 109. Through these two data, the recognition recall rate of the software for the category label of apple(scab) can be obtained, resulting in a

ratio of 109 and 125, giving 0.87. Similarly, dividing all the element values on the main diagonal of the confusion matrix by the sum of the column or row element values can obtain the accuracy or recall rate of the corresponding row label, as shown in Fig. 7. It can be seen that the accuracy rate is similar to the recall rate in most categories, but the recall rate is much higher than the accuracy rate in grape black rot, indicating that although the model cannot accurately identify this category, it can separate the disease from other conditions in a large range. Therefore, the model is still effective for the identification of this kind of disease, which can help agricultural workers find this kind of disease.

In addition, 29 pieces of grape (leaf blight) were recognized as grape(black rot). The analysis shows that grape (black rot) is reddish-brown circular or oval lesions, while grape (leaf blight) lesions are similar in shape and color, so the identification of the two is easy to confuse, so the recognition accuracy is low. The identification results of the other 6 kinds of pests and health categories are relatively accurate. It can be seen that the model has a strong ability to identify leaves with pests and diseases, and can be applied in complex natural environments.

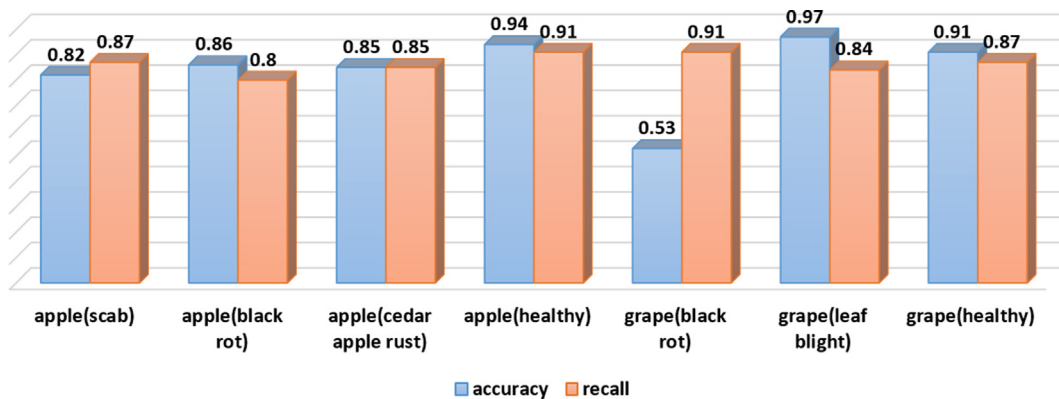


Fig. 7 – The accuracy and regression rate of the adopted model for different categories on the test set.

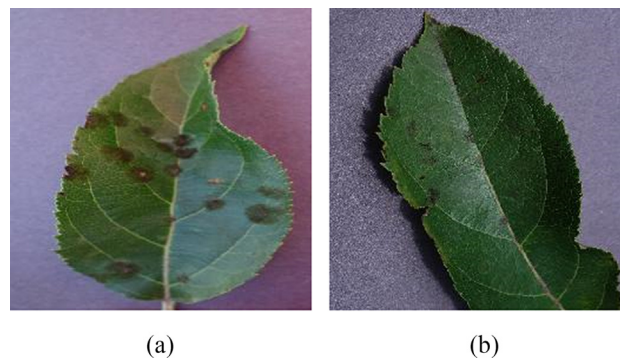


Fig. 8 – Example of the classification result of scab image: (a) Correct sample; (b) Error sample.

The Figs. 8-14 show the correct and error examples of each category predicted by the model. It can be found that for the samples with obvious characteristics, the model can accurately predict, while for some samples with less obvious characteristics. The main reason for the error classification is that these disease characteristics are close to each other, or there are few errors in the training set, leading to errors in the pre-

diction process of the model. It affects the classification accuracy of the model.

As shown in Fig. 15 and Fig. 16, the accuracy and loss show upward and downward convergence, respectively, during the training of the model, which indicates that the model gradually learns to distinguish samples from different categories. In the process of training the model, both train_acc and vali-

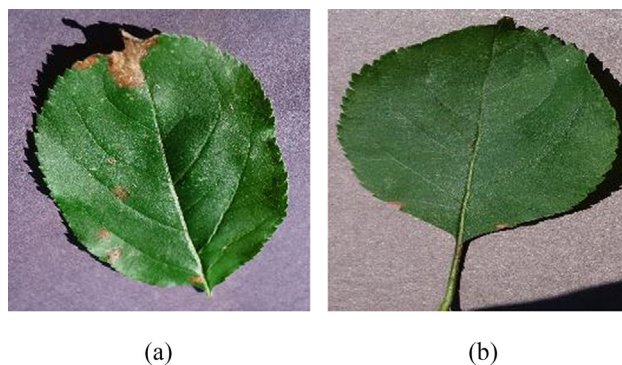


Fig. 9 – Example of the classification result of apple black rot image: (a) Correct sample; (b) Error sample.

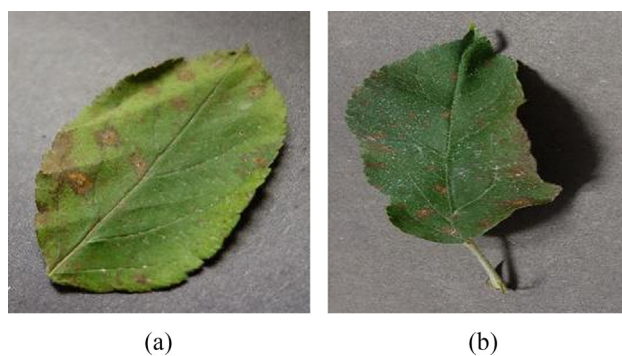


Fig. 10 – Example of the classification result of cedar apple rust image: (a) Correct sample; (b) Error sample.

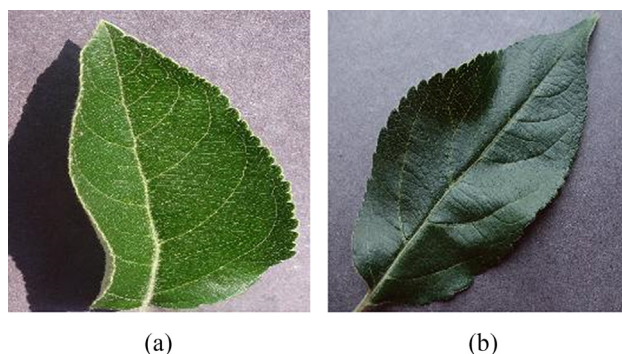


Fig. 11 – Example of the classification result of apple healthy image: (a) Correct sample; (b) Error sample.

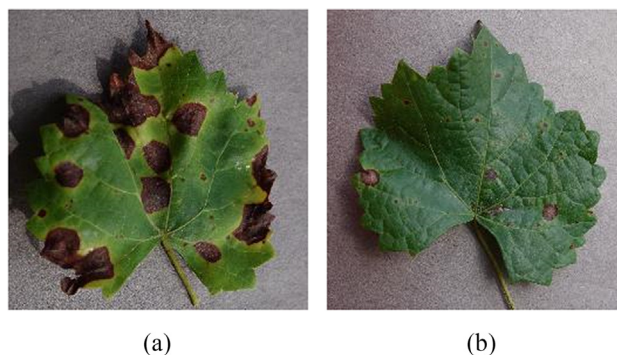


Fig. 12 – Example of the classification result of grape black rot image: (a) Correct sample; (b) Error sample.

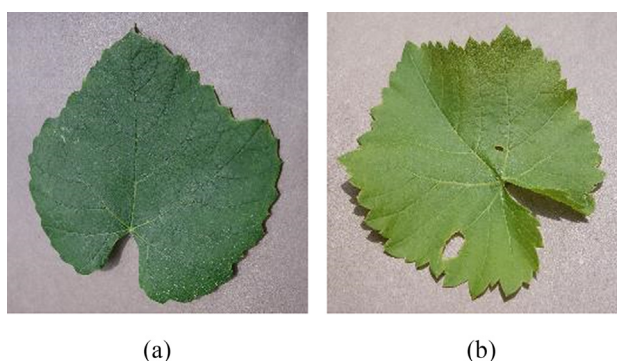


Fig. 13 – Example of the classification result grape healthy image: (a) Correct sample; (b) Error sample.

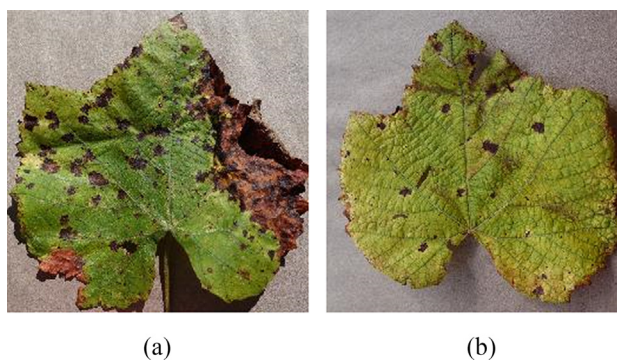


Fig. 14 – Example of the classification result leaf blight image: (a) Correct sample; (b) Error sample.

d_acc gradually increased, and then stabilized after maintaining convergence, among which train_acc was generally slightly higher than valid_acc. In addition, both train_loss and valid_loss gradually decreased and stabilized after maintaining convergence, among which train_loss was slightly lower than valid_loss overall.

3.2. Discussion

To achieve fast and accurate recognition of crop leaf diseases, a neural network model based on improved ViT is proposed to recognize leaf diseases in complex environments in the field.

The visual attention mechanism refers to the mechanism by which the human brain focuses its attention on the endpoint area of interest when observing the target thing visually, and in the case of leaf surface disease pictures, on the disease features. Similarly, the attention mechanism implemented by the neural network algorithm can make the neural network model focus more on the important parts of the picture and ignore the secondary parts. Therefore, the ViT-based model proposed in this paper uses the transformer as the attention mechanism, thus focusing on the features that reflect plant diseases in the leaf surface. In addition, a chunking method is used to select the most distinctive regions in the image,

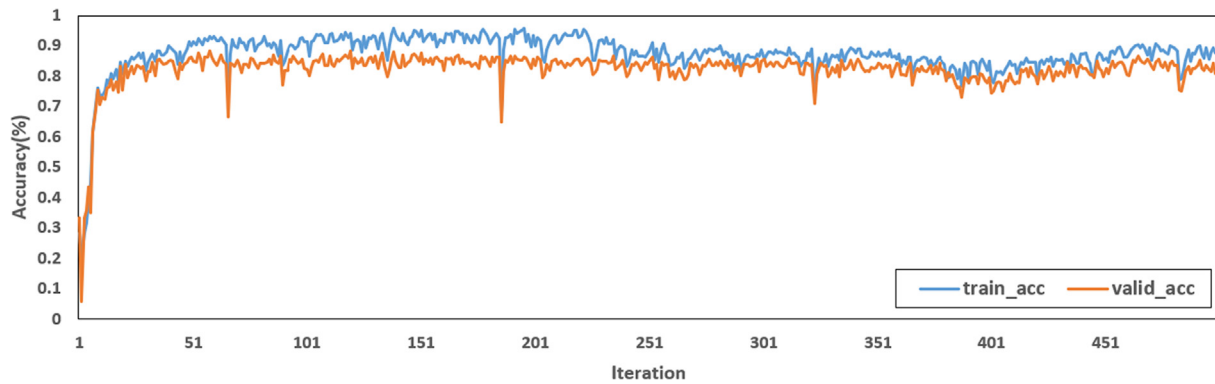


Fig. 15 – Model training accuracy iteration curve.

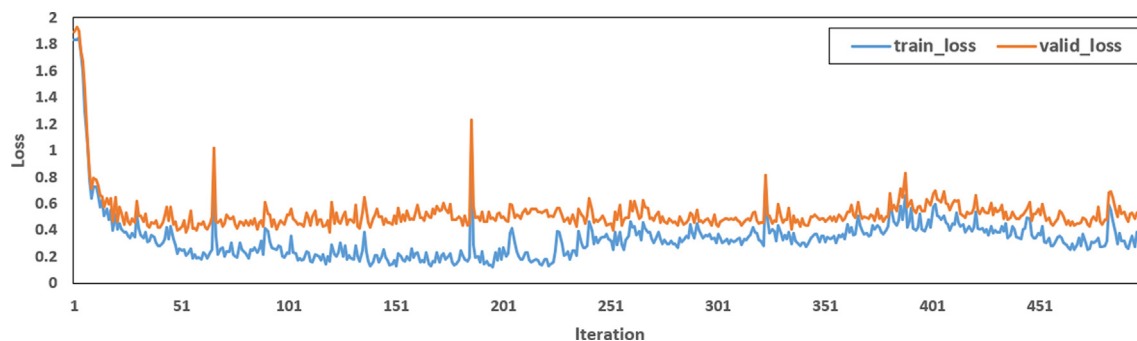


Fig. 16 – Model training loss function iteration curve.

which in turn can better exploit the accuracy of pest and disease features. At present, scholars have applied neural networks to recognize leaf disease images and achieved better results. However, there are some drawbacks, such as simple background and a limited number of categories leading to poor generalization ability. The proposed improved ViT-based model showed high accuracy in experiments for both grape leaf and apple leaf disease detection.

However, since the model is obtained based on ViT model, it is inevitable that it suffers from some defects inherent to the ViT model. ViT slices the input image into image patches. The rules of this slicing method are simple, fixed, and mandatory, which may destroy the spatial structure of the image. The model will ignore the spatial information, especially the edge information of between patches, thus it is easy to ignore some plant disease feature information such as the tiny feature and the edge feature between patches. In addition, the training cost of the model increases due to the high computational complexity of the self-attention mechanism. To improve the above flaws, we consider introducing a local attention mechanism in the subsequent study, which extracts the local features of the image and performs local and global feature interaction. The high training cost of ViT model is also reflected in the amount of data used to train the model, which is due to weak inductive bias capability of ViT, so it can only obtain better performance with pre-training of large-scale datasets, and when the training it is difficult to exploit its

advantages when the training data or the computational power of the hardware is limited. In view of this problem, we consider incorporating CNN (Convolutional Neural Networks) into ViT-based pest identification models in subsequent studies, relying on the inductive bias capability of CNN to reduce the number of training datasets required by the models and thus reduce the cost of training models. It can be seen from the experimental results that the imbalance of the training dataset leads to the model having poor accuracy for a small number of categories, for example, only 0.53 accuracy for the classification of grape leaves suffering from black rot, but high accuracy for the classification of cedar apple rust and leaf blight with the sufficient number of samples. Therefore, we will also focus on the problem of data imbalance in our subsequent research work to improve the accuracy of disease detection as a whole.

Overall, the model has a good overall performance and high recognition accuracy. Therefore, this work has practical application value. In the future, we will continue our in-depth research work on disease detection and hope to contribute to the development of smart agriculture.

4. Conclusions

In order to realize effective and efficient classification of crop diseases, we introduced a model based on improved Vision Transformer. The model with self-attention mechanism can

focus on the most important parts of the picture, which reflected the plant disease characteristics in the leaves. In addition, the segmentation method was used to select the most obvious feature area in the image, so as to better excavate the feature accuracy of pests and diseases. Through experiments, we found that the method showed good results in the detection of diseases, indicating that the self-attention mechanism can optimize the performance of the method in crop diseases classification. In the future, we will further study the existing defects of this model in the discussion section.

Availability of data and materials

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

CRedit authorship contribution statement

Xueqian Fu: Conceptualization, Methodology, Writing – review & editing, Supervision. **Qiaoyu Ma:** Methodology, Writing – original draft. **Feifei Yang:** Investigation, Writing – original draft. **Chunyu Zhang:** Formal analysis. **Xiaolong Zhao:** Formal analysis. **Fuhao Chang:** Methodology. **Lingling Han:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study is supported by the National Natural Science Foundation of China under Grant 52007193 and The 2115 Talent Development Program of China Agricultural University.

REFERENCES

- [1] Zhaoyu Zhai, Cao Xu, Huanliang Yifei, et al. Review of Key Techniques for Crop Disease and Pest Detection. *Trans Chin Soc Agric Machin* 2021;52(7):1–18. <https://doi.org/10.6041/j.issn.1000-1298.2021.07.001>.
- [2] Fu X, Guo Q, Sun H. Statistical machine learning model for stochastic optimal planning of distribution networks considering a dynamic correlation and dimension reduction [J]. *IEEE Trans Smart Grid* 2020;11(4):2904–17.
- [3] Fu X. Statistical machine learning model for capacitor planning considering uncertainties in photovoltaic power[J]. *Protect Control Modern Power Syst* 2022;7(1):5.
- [4] Liu J, Wang X. Plant diseases and pests detection based on deep learning: a review[J]. *Plant Methods* 2021;17:1–18.
- [5] Ngugi LC, Abelwahab M, Abo-Zahhad M. Recent advances in image processing techniques for automated leaf pest and disease recognition—A review[J]. *Inform Proc Agric* 2021;8(1):27–51.
- [6] Fu X, Zhou Y. Collaborative Optimization of PV Greenhouses and Clean Energy Systems in Rural Areas. *IEEE Transactions on Sustainable Energy* 10.1109/TSTE.2022.3223684.
- [7] Xueqian Fu, Niu H. Key Technologies and Applications of Agricultural Energy Internet for Agricultural Planting and Fisheries industry. *Inform Process Agric* 2022. <https://doi.org/10.1016/j.inpa.2022.10.004>.
- [8] Zhang J, Huang Y, Ruiliang Pu, et al. Monitoring plant diseases and pests through remote sensing technology: A review. *Comput Electron Agric* 2019;165. <https://doi.org/10.1016/j.compag.2019.104943> 104943.
- [9] Dan B, Sun X, Liu L. Diseases and Pests Identification of Lycium Barbarum Using SE-MobileNet V2 Algorithm. In: 2019 12th International Symposium on Computational Intelligence and Design (ISCID). p. 121–5. <https://doi.org/10.1109/ISCID.2019.00034>.
- [10] Tammy S, Boggess SL, Denita H, et al. Conventional Gel Electrophoresis and TaqMan Probes Enable Rapid Confirmation of Thousand Cankers Disease from Diagnostic Samples. *Plant Dis* 2021;10(105):3171–80. <https://doi.org/10.1094/PDIS-10-20-2258-RE>.
- [11] Xie C, Wang R, Zhang J, et al. Multi-level learning features for automatic classification of field crop pests. *Comput Electron Agric* 2018;152:233–41. <https://doi.org/10.1016/j.compag.2018.07.014>.
- [12] Ngugi C, Lawrence, Moataz, Abelwahab, Mohammed, Abo-Zahhad. Recent advances in image processing techniques for automated leaf pest and disease recognition – A review. *Inform Proc Agric* 2021;8(1):27–51. <https://doi.org/10.1016/j.inpa.2020.04.004>.
- [13] Kale MRSR, Shitole MRS. Analysis of Crop disease detection with SVM, KNN and Random forest classification. *Inform Technol Ind* 2021;1(9):364–72. <https://doi.org/10.17762/iti.v9i1.140>.
- [14] Gowrishankar K, LakshmiPraba S, Darshini V, Divya. A Hybrid Model for Pest identification in Groundnut Plants using BLR and SVM Techniques. [J]. *Ann Roman Soc Cell Biol* 2021:12363–71.
- [15] Raza Shan-e-Ahmed, Prince Gillian, Clarkson John P., et al. Automatic Detection of Diseased Tomato Plants Using Thermal and Stereo Visible Light Images[J]. *PloS one*, 2015 4(10):e123262. doi: 10.1371/journal.pone.0123262.eCollection 2015.
- [16] Wang K, Chen K, Huiyu Du, et al. New image dataset and new negative sample judgment method for crop pest recognition based on deep learning models. *Eco Inform* 2022;69. <https://doi.org/10.1016/j.ecoinf.2022.101620> 101620.
- [17] Wang R, Jiao L, Xie C, et al. S-RPN: Sampling-balanced region proposal network for small crop pest detection. *Comput Electron Agric* 2021;187. <https://doi.org/10.1016/j.compag.2021.106290> 106290.
- [18] Lucas M. Tassis, João E. Tozzi de Souza, et al., A deep learning approach combining instance and semantic segmentation to identify diseases and pests of coffee leaves from in-field images, *Computers and Electronics in Agriculture*, Volume 186, 2021, 106191, doi: 10.1016/j.compag.2021.106191.
- [19] Yang J, Lan G, Li Y, et al. Data quality assessment and analysis for pest identification in smart agriculture. *Comput Electr Eng* 2022;103. <https://doi.org/10.1016/j.compeleceng.2022.108322> 108322.
- [20] Jackulin C, Murugavalli S, A comprehensive review on detection of plant disease using machine learning and deep learning approaches, *Measurement: Sensors*, Vol. 24, 2022, 100441, doi: 10.1016/j.measen.2022.100441.
- [21] Fu, Zujun, Transformer in Computer Vision: ViT and its Progress. arXiv preprint arXiv:2205.11239; 2022.